

MANAV GARG

Sonipat, Delhi NCR, India

+91 7837139775 manavgarg2326@gmail.com linkedin.com/in/manav-garg-4a9264377 github.com/manav363

Education

Rishihood University

Bachelor of Technology in Computer Science & AI

Sep. 2025 – May 2029

Sonipat, Delhi NCR

Projects

Market Regime Detection | Python, FastAPI, Scikit-learn, Pandas, Joblib 2026

- Built a quantitative research system for Indian equity markets (NSE/NIFTY) covering the full pipeline — data loading, feature engineering, walk-forward model training, signal generation, risk management, and backtesting across multiple market regimes.
- Engineered out-of-sample (OOS) predictions explicitly into the modeling path to prevent data leakage; structured the codebase as a composable research pipeline via `pipeline.py` rather than notebook sprawl, enabling reproducible experimentation.
- Achieved 115/115 tests passing across all 7 development phases with systematic phase-gate validation; built a Bloomberg Terminal-style dark dashboard with Analyze, Scanner, Backtest, and Risk tabs over a FastAPI backend with real-time signal updates.

APIBlueprint | React, Vite, FastAPI, SQLAlchemy, PostgreSQL, Express, Docker Compose 2026

- Designed and built a full-stack API design studio where users define REST endpoints, auto-generate OpenAPI specs, and spin up a dynamic mock server — all from a single project data model.
- Implemented a coherent data loop: CRUD persistence via SQLAlchemy → OpenAPI spec generation → Express mock server synthesizing JSON responses from schemas → live docs and export.
- Secured the tool with HTTP Basic auth shared across the backend and mock server, with a React frontend login gate, ErrorBoundary, and Alembic-managed database migrations.
- Designed the mock server to synthesize contextually realistic JSON responses from OpenAPI schemas, enabling frontend teams to develop against live-feeling endpoints before backend implementation is complete.

SentiScope | Python, FastAPI, HuggingFace Transformers, React, TypeScript, Upstash Redis, Docker 2026

- Built a full-stack sentiment analysis app accepting pasted text or article URLs, returning positive/neutral/negative classifications with confidence breakdowns via a React dashboard.
- Implemented SSRF protection on URL ingestion, request correlation via `X-Request-ID`, chunked text inference with HuggingFace pipelines, and Upstash Redis caching — the best-tested project in the portfolio.
- Containerized backend and frontend with separate Dockerfiles; backend pytest suite passes end to end across text and URL analysis flows.
- Designed the system to handle edge cases including malformed URLs, oversized payloads, and ambiguous text encoding; implemented structured error responses and graceful degradation across all inference failure modes.

CineStats | Vanilla JavaScript, HTML/CSS, TMDB API 2026

- Built a premium-feeling movie discovery dashboard in pure vanilla JavaScript — no frameworks — with modular separation across `api.js`, `filters.js`, `ui.js`, and `app.js`.
- Implemented dynamic filtering, search, and sort logic against the TMDB API with clean DOM rendering and polished visual presentation, demonstrating frontend fundamentals without framework dependency.
- Optimized API call efficiency with debounced search inputs and client-side caching to minimize redundant TMDB requests, resulting in a snappy user experience without a dedicated backend layer.

Technical Skills

Languages: Python, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: FastAPI, Pandas, Scikit-learn, SQLAlchemy

Databases & Storage: PostgreSQL, Redis, DuckDB

Tools & Platforms: Docker, Git, GitHub, JWT, REST APIs, Linux

Architecture Patterns: Microservices, Circuit Breaker, Event-Driven Design, Test-Driven Development

Domain Knowledge: Algorithmic Trading, NSE/NIFTY Markets, Financial Data Pipelines, Market Regime Analysis

Relevant Coursework

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|-----------------------|---------------------------|------------------------|-------------------------|
| • Data Structures | • Database Management | • Systems Programming | • Computer Architecture |
| • Algorithms Analysis | • Artificial Intelligence | • Software Engineering | • Internet Technology |